

HarmonyXR Gaze-Pose Context System

Project Completion and Status Report

Report Date: May 20, 2026

Project: HarmonyXR Gaze-Pose Context System

Prototype: Adaptive XR Context Prototype

Current Branch: research/harmonyxr/unity-6.4-migration

Primary Scope: Unity 6 XR proof-of-concept for context inference, adaptive guidance, Quest execution, headset evidence, and research communication.

1. Executive Summary

The HarmonyXR Gaze-Pose Context proof-of-concept has progressed from a technical Unity XR prototype into a polished, research-driven adaptive XR demonstration that has now been migrated and verified on Unity 6000.4.7f1. The current system runs as a Quest-based object-sorting experience and communicates the research purpose through onboarding, live guidance, adaptive support messages, runtime QA evidence, app-generated context logs, and a completion summary.

The prototype demonstrates that multimodal XR signals can be captured during a task and used to infer user context states such as Engaged, Distracted, Transitioning, and Idle. Based on these inferred states, the interface adapts user-facing guidance and support without changing the approved research scope or adding unrelated gameplay systems. The May 20 Quest QA run produced a valid app context log with 2,220 records and all four context states.

The current project is ready for internal review, stakeholder demonstration, and paper-supporting evidence use. Remaining work is limited to final visual evidence capture, manual QA note completion, optional demo video capture, and any supervisor-requested documentation adjustments.

2. Project Objective

The main objective of this work was to improve the Unity XR proof-of-concept so first-time users, clients, supervisors, and stakeholders can clearly understand:

- what the application is,
- why the user is performing the task,
- what the system is detecting,

- what research problem is being addressed,
- and why adaptive XR behavior matters.

The work stayed within the approved proof-of-concept boundary. No new gameplay loop, unrelated study system, mobile-phone mode, or expanded research architecture was added.

3. Initial Project State

At the start of this improvement cycle, the core technical pipeline already existed:

- TrainingSimulation object-sorting task,
- context inference pipeline,
- four context states: Engaged, Distracted, Transitioning, and Idle,
- user guide panel,
- adaptive behavior support,
- debug/QA overlay,
- context logging.

However, the experience still felt closer to a technical Unity prototype than a polished research demonstration. The main gaps were communication, readability, stakeholder confidence, and presentation quality.

Key issues identified:

- onboarding explained the task more than the research purpose,
 - users could understand sorting but not the value of multimodal context inference,
 - UI panels were visually dense or debug-like,
 - completion feedback did not strongly communicate research success,
 - QA evidence was useful but needed clearer export and interpretation,
 - headset demo visuals were still plain due to the default skybox.
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4. Work Completed

4.1 Onboarding and Research Communication

The onboarding flow was improved to explain the application as an adaptive XR research prototype rather than only a sorting task.

Completed improvements:

- added clearer research-purpose language,

- improved explanation of the four context states,
- refined instructional wording for professional tone,
- made onboarding progression user-controlled,
- reduced dense paragraph-style guidance,
- improved visual hierarchy and spacing,
- aligned the guide panel more comfortably for headset viewing.

Current result:

The user is introduced to the experience step by step and can understand that the sorting task is used to create observable behavior for context inference.

4.2 Adaptive XR Guidance

The prototype now communicates adaptive behavior more clearly during runtime.

Implemented adaptive behavior communication:

- Engaged: continues normal task guidance when the user appears focused,
- Distracted: presents refocus or attention guidance,
- Transitioning: gives soft guidance while the user moves between task steps or targets,
- Idle: presents adaptive support actions such as Continue, Take a break, and Recenter view,
- Incorrect selection: shows corrective guidance when the wrong object is selected,
- Completion: presents a research-focused summary after all objects are sorted.

Current result:

The adaptive behavior is visible to users and stakeholders as runtime guidance and support, not only as hidden internal state logic.

4.3 Object Task Feedback

The object-sorting task was refined so the user receives clearer feedback when the active object does not match the selected object.

Completed improvements:

- added incorrect-object selection messaging,
- reinforced the active task object,
- helped users recover without changing the task design,
- preserved the simple research-controlled interaction flow.

Current result:

The task is easier to follow, and incorrect selections now support the research narrative by showing context-aware guidance.

4.4 QA and Runtime Evidence Logging

Runtime QA evidence was improved so important context values can be reviewed from headset console logs.

Captured QA fields include:

- STATE,
- CONF,
- BOUND,
- AOI,
- POSTURE,
- FIX,
- DWELL,
- SACC,
- BODY,
- HAND,
- PINCH,
- DIST.

Current result:

The system produces runtime QA evidence that supports research validation and explanation of the context inference pipeline.

4.5 QA Panel and Developer Presentation

The QA/Event panel behavior was refined to reduce the debug-build feeling during normal user interaction.

Completed improvements:

- avoided showing the QA panel for simple pinch actions alone,
- repositioned the panel away from the main interaction area,
- improved terminology from developer/debug language toward research-oriented language,
- kept QA information available for evidence and developer explanation.

Current result:

The normal user experience is cleaner, while QA mode remains useful for demonstrating internal evidence when required.

4.6 End-of-Session Completion Experience

The final completion screen was redesigned to make the result clearer and more professional.

Completed improvements:

- improved spacing and hierarchy,
- added stronger task-completion messaging,
- removed repeated or merged text,
- highlighted adaptive XR context analysis completion,
- presented a clearer research proof statement.

Current result:

The end of the experience now communicates that the task was completed and that adaptive XR context analysis was demonstrated.

4.7 Headset Evidence Collection

A combined headset evidence run was completed and organized on May 14, 2026. A second Unity 6 Quest QA evidence run was completed on May 20, 2026 after migration to Unity 6000.4.7f1.

Evidence collected:

- May 14 headset evidence video and processed QA tables,
- May 20 ADB device and package evidence,
- May 20 Android activity dump confirming UnityPlayerGameActivity,
- May 20 Unity runtime logcat evidence,
- May 20 error-only logcat evidence,
- May 20 memory, graphics, and thermal dumps,
- May 20 app-generated context JSONL output,
- May 20 parsed context-state counts and summary JSON.

Verified May 20 context-state coverage from 16_context_log_20260520_115419.jsonl:

State	Observed	Count
Idle	Yes	926
Distracted	Yes	620
Transitioning	Yes	530
Engaged	Yes	144

Current result:

The evidence package supports the claim that all four context states were observed during headset execution. The current Unity 6 run produced 2,220 valid context JSONL records.

4.8 Research Paper Draft

A Lavanya-focused research paper draft was prepared using current implementation details and collected evidence.

Completed outputs:

- Markdown paper source,
- DOCX paper draft,
- PDF paper draft,
- supporting figures,
- Phase 4/user study treated only as future work.

Current result:

The paper draft is available for review and further refinement. It does not claim completed user-study results.

4.9 Demo Presentation Preparation

The project was prepared for presentation through script and visual-polish work.

Completed improvements:

- prepared a LinkedIn/company demo voiceover script,
- aligned the script with the actual headset flow,
- described onboarding, task execution, incorrect selection, adaptive support, evidence logging, and completion,
- added a custom neutral demo skybox to replace the Unity default skybox in the TrainingSimulation scene.

Current result:

The demo is better prepared for video recording and stakeholder presentation. The new skybox should be visually checked in Unity/headset before final sharing.

5. Evidence Files

Primary evidence files currently available:

- May 14 full headset evidence video:
Research_Evidence/03_Context_State_Coverage/20260514_run01_full-evidence.mp4

- May 14 console and filtered evidence:
Research_Evidence/05_Stability_Latency/
- May 14 processed QA tables: Research_Evidence/07_Processed_Tables_Charts/
- May 20 QA index:
Research_Evidence/qa_quest_test_20260520/00_qa_evidence_index.json
- May 20 ADB/device/package/activity evidence: 01_adb_devices.txt through
07_activity_top_after_launch.txt
- May 20 Unity runtime logcat: 08_runtime_logcat.txt
- May 20 error-only logcat: 09_errors_only_logcat.txt
- May 20 memory, graphics, and thermal evidence: 12_meminfo.txt, 13_gfxinfo.txt,
14_thermal_status.txt
- May 20 app-generated current context log:
16_context_log_20260520_115419.jsonl
- May 20 legacy context log: 17_legacy_context_log.json
- May 20 parsed state counts: 18_context_state_counts.csv
- May 20 parsed context summary: 19_context_log_summary.json

These files are stored under the Research_Evidence folder using the organized evidence structure created for the project. The May 20 screenshot and ADB screenrecord files are currently zero-byte files due to Quest/XR layer capture limitations and should not be used as visual proof.

6. Current Project Status

Current status:

The approved Unity XR proof-of-concept implementation is functionally complete for the current research-demonstration scope and is ready for review, paper-supporting use, and final demo capture.

Reason for this status:

- the headset experience runs and supports the sorting task on Quest 2,
- the project has been migrated to Unity 6000.4.7f1,
- the Android activity entry now uses Unity 6's UnityPlayerGameActivity,
- onboarding and guidance now communicate the research purpose,
- the system demonstrates four context states,
- adaptive guidance is visible during runtime,
- incorrect selection and idle support behavior are represented,
- completion screen communicates research success,
- QA metrics and context logs are captured from the headset,

- all four context states are present in the May 20 app-generated context JSONL evidence,
- paper and report sources are updated for the Unity 6 Quest QA state,
- visual presentation has been improved with a custom demo skybox.

This status does not mean the full research program is complete. It means the current proof-of-concept and evidence package are complete enough for supervisor review, stakeholder demonstration, and paper-draft support.

7. Remaining Work

Remaining work is limited to review, validation, and final presentation tasks:

- create `15_manual_qa_notes.txt` with headset-observed pass/fail results,
 - recapture visual proof using Quest screenshot, external phone photo, or external video because ADB screenrecord returned an XR layer-stack error,
 - record the final demo video using the prepared script if a demo video is required,
 - review the paper draft with supervisor/team feedback,
 - update the paper if requested,
 - avoid claiming Phase 4 user-study outcomes until that work is actually conducted,
 - keep evidence files organized for final submission and presentation.
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8. Important Limitations

The following limitations should be stated clearly when presenting the project:

- This is a proof-of-concept, not a completed large-scale user study.
 - Phase 4/user-study work is future work unless separately completed.
 - The adaptive behavior is intentionally lightweight and focused on guidance/support.
 - The system adapts UI guidance and feedback; it does not introduce unrelated gameplay or autonomous task redesign.
 - `room_scale` posture values should be interpreted as spatial/boundary context, not strict sitting/standing classification.
 - The current evidence supports runtime state coverage and signal logging, but formal statistical user validation remains outside the completed scope.
 - ADB screenrecord did not produce usable visual evidence in the May 20 Quest run, so visual proof should come from a headset screenshot recapture or external camera recording.
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9. Final Assessment

The project has successfully moved from a technically working Unity XR prototype to a clearer, more presentable adaptive XR research proof-of-concept that is now verified on Unity 6 and Quest 2.

The current implementation demonstrates:

- headset-based task execution,
- multimodal XR signal capture,
- context-state inference,
- adaptive runtime guidance,
- corrective task feedback,
- idle support behavior,
- research-focused completion summary,
- QA evidence logging,
- organized evidence package,
- Unity 6 migration and Quest launch verification,
- app-generated context JSONL output with all four states,
- and presentation-ready documentation outputs.

The project is currently suitable for final demo preparation, internal team review, stakeholder explanation, and research-paper refinement within the approved scope.

10. Unity 6 Migration and Quest Verification Update - May 20, 2026

After the May 15 report, the Unity project was migrated and verified on Unity 6000.4.7f1.

Completed update work:

- stabilized the Unity project after package import and Safe Mode issues,
- confirmed the TrainingSimulation scene and project assets were preserved,
- replaced Meta XR All-in-One with Meta XR Core to avoid duplicate Oculus Android namespace conflicts,
- restored required direct dependencies including Input System and TextMeshPro,
- applied Meta XR Project Setup Tool fixes for Quest/Android compatibility,
- updated the Android manifest to use Unity 6's `com.unity3d.player.UnityPlayerGameActivity`,
- fixed the Quest loading-screen issue caused by the invalid `AppUIGameActivity` entry,

- added a camera fallback in XRAppShellController so the scene retains a valid render camera if rig bootstrapping is incomplete,
- replaced obsolete project-script API calls for Unity 6 compatibility, including object lookup and TextMeshPro wrapping APIs,
- built, installed, launched, and verified the APK successfully on Meta Quest 2,
- collected ADB, logcat, package/activity, memory, graphics, thermal, and app context-log evidence.

Current verified environment:

- Unity Editor: 6000.4.7f1
- OpenXR Plugin: 1.16.1
- Meta XR Core SDK: 201.0.0
- URP: 17.4.0
- Target device: Meta Quest 2
- Runtime scene: Assets/Scenes/TrainingSimulation.unity
- Android package: com.UnityTechnologies.com.unity.template.urpblank
- Runtime activity: com.unity3d.player.UnityPlayerGameActivity
- Quest device ID used for QA: 1WMHH836GS1082
- Device OS evidence: Android 14, SDK 34

May 20 evidence summary:

Evidence Item	Result
ADB device connection	Captured
Package installed	Captured
Unity activity running	UnityPlayerGameActivity confirmed
Runtime logcat	Captured, 27,021 lines
Current context JSONL	Captured, 2,220 valid records
Context states observed	Idle, Distracted, Transitioning, Engaged
Memory/gfx/thermal dumps	Captured
Screenshot/video	Needs visual recapture; ADB media files were zero bytes

Current status:

The application now launches correctly on Quest 2 and the main simulation flow is working. The paper and report are updated with the current Unity 6 Quest QA evidence. Remaining work is limited to manual QA notes, visual proof recapture, optional demo video capture, APK packaging if needed, and final repository backup.